Same structure, different stability: The lasting legacy of modelling assumptions

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Abstract: Food web reconstruction relies on generative models that embody different assumptions about how trophic interactions arise. Yet whether these alternative representations provide interchangeable starting points for predicting ecosystem dynamics remains unclear. Here, we compared seven food web generative models spanning structural, statistical and trait-based approaches, embedding each within a common bioenergetic simulation framework while controlling species pools and dynamical rules. Initially, networks generated by different models occupied distinct regions of multivariate topology space. Following bioenergetic simulations, however, realised networks converged towards a shared region through extinction-driven pruning of species and interactions, suggesting that ecological dynamics constrain the structures of persistent communities. This structural convergence did not translate into equivalent dynamical predictions. Equilibrium biomass predictions remained model-dependent, as did local stability properties, including resilience and reactivity. By contrast, variation in several broader network and aggregate properties was explained primarily by realised species richness rather than reconstruction framework. These contrasting patterns suggest that different aspects of ecosystem behaviour depend on different levels of ecological organisation and on the information retained during network reconstruction. In particular, models differ in whether interactions are constrained by trophic ordering, organismal traits and energetic relationships, or network-level properties. Our findings demonstrate that convergence in food web topology does not necessarily imply functional or dynamical equivalence. Food web models should therefore be evaluated not only by their ability to reproduce observed network structure, but also by whether the ecological assumptions and information they encode are appropriate for predicting the particular aspects of ecosystem functioning under investigation.

Keywords: food web, bioenergetic dynamics, network stability, ecological networks, generative models, community persistence

1 Introduction

For over two decades, community ecology has relied heavily on generative models to understand and reconstruct the structure of trophic networks. At the centre of this framework is the niche model. Introduced by Williams & Martinez (2000) as a major advance over the strictly hierarchical Cascade model (Cohen et al., 1990), the Niche model demonstrated that relatively simple assembly rules could reproduce many emergent properties of empirical food webs, including realistic degree distributions, trophic guild proportions, and path lengths. This success has established the niche model as the default structural baseline for investigating macroecological patterns, robustness to primary extinctions, and the stability of ecological communities (Allesina & Tang, 2012; Curtsdotter et al., 2011).

The widespread use of the Niche model has also encouraged a broader assumption that different food web reconstruction methods provide interchangeable representations of ecological communities. However, generative models embody fundamentally different assumptions about how trophic interactions arise (Strydom, Dunhill, et al., 2026). Structural models operate from the network down, generating interactions according to statistical or topological rules. The Niche and Cascade models impose ordered feeding hierarchies along a latent niche axis, while maximum entropy (MaxEnt) models generate networks that satisfy macroscopic constraints while remaining maximally unstructured beyond those constraints (Banville et al., 2023). In contrast, mechanistic approaches operate from the species up, constructing networks from organismal traits and ecological processes. The Allometric Diet Breadth Model (ADBM) derives realised diets from optimal foraging and energetic profitability (Petchey et al., 2008), whereas the Allometric Trophic Network (ATN) framework constrains interactions through body-size scaling and metabolic theory (Brose et al., 2006; Schneider et al., 2016). Latent trait models occupy an intermediate position, inferring interactions statistically from underlying species traits without prescribing explicit feeding rules (Rohr et al., 2010).

These contrasting modelling philosophies raise a fundamental question. Are they simply different routes to the same ecological inference, or do they encode different assumptions about network architecture that persist into predictions of ecosystem dynamics? Recent work has shown that applying different reconstruction frameworks to identical species pools produces markedly different food web topologies and inference about species extinctions (Strydom, Karapunar, et al., 2026). Because network architecture governs the pathways

through which energy, biomass, and indirect effects propagate through communities (Delmas et al., n.d.), differences in reconstruction may have important consequences for ecological dynamics rather than representing alternative but equivalent descriptions of the same system.

At the same time, empirical food webs exhibit surprisingly consistent local structural organisation despite large differences in species composition and ecosystem type, suggesting that ecological communities occupy a constrained region of possible network architectures rather than arbitrary combinations of interactions. Understanding how this organisation relates to ecological stability has been a longstanding ambition in ecology, underpinning the diversity–stability debate and efforts to identify the structural properties that allow complex communities to persist (McCann, 2000; Thompson et al., 2012). This has motivated the idea that ecological dynamics selectively retain interaction structures that are compatible with persistence while filtering those that are not (Stouffer et al., 2007; Stouffer & Bascompte, 2010). Whether this filtering is sufficiently strong to erase the structural differences introduced by alternative reconstruction frameworks remains unknown.

Here, we test whether food web reconstruction frameworks function as benign structural templates or whether they fundamentally shape ecological inference. We generated food webs using seven models spanning structural, statistical, and trait-based philosophies and embedded each within an identical bioenergetic simulation framework while controlling species pools and dynamical rules. This design allows us to separate the effects of network construction from the effects of ecological dynamics and ask two questions: (i) do different reconstruction frameworks converge towards a common realised network structure under bioenergetic dynamics, and (ii) if structural convergence occurs, do predictions of biomass equilibrium also converge, or do they retain a signature of the original generative model?

2 Methods

2.1 Network generation

We generated food webs using six established generative models, spanning structural (Niche, Cascade, Random, MaxEnt) and trait-based approaches (Latent Trait Model, Allometric Trophic Network, Allometric Diet Breadth Model), see Table 1 for more details. All simulations were implemented in Julia (Bezanson et al., 2017) and analyses in R (R

Core Team, 2024).

Table 1: Summary of food web models used in this study. Structural models (Niche, Cascade, Random, MaxEnt) generate network topology based on abstract or statistical rules, while realised/dynamic models (ADBM, ATN, Latent trait) incorporate species-specific traits and biological mechanisms to predict interactions. For each model, we list the main inputs and the key assumptions that govern link formation.

Model	Type	Main Inputs	Key Assumptions
Niche model (Williams & Martinez, 2000)	Structural	Species niche values	Trophic interactions structured by a one-dimensional feeding niche; species consume all prey within a contiguous range; allows cannibalism.
Cascade model (Cohen et al., 1990)	Structural	Species rank	Consumers feed only on species with lower ranks; strictly hierarchical; no loops or omnivory.
MaxEnt model (Banville et al., 2023)	Structural	Species richness, connectance	Links assigned to maximise entropy; captures global network properties without species-specific mechanisms.
Random model (Erdős & Rényi, 1959)	Structural	Species richness, connectance	Based on the Erdős–Rényi model, links are assigned randomly between nodes

Model	Type	Main Inputs	Key Assumptions
Allometric Diet Breadth Model (ADBM) (Petchey et al., 2008)	Realised	Body mass, prey energy content	Consumers maximize energy intake; diet breadth determined by profitability and handling time; allometric scaling governs parameters.
Allometric Trophic Network (ATN) (Brose et al., 2006)	Realised	Body mass	Interactions constrained by body mass ratios; mechanical size limits structure networks; probability of interaction follows Ricker function.
Latent trait model (Rohr et al., 2010)	Realised	Species traits (*e.g.,* body mass)	Interactions inferred statistically based on hidden trait correlations; can incorporate probabilistic constraints on links.

To isolate model effects from input variation, all trait-based models were parameterised using an identical species pool at each replicate. Species richness was set to 10, 20, 40, and for each iteration we sampled (i) a target basal fraction (U(0.1,0.3)), (ii) metabolic classes consistent with this fraction, and (iii) body masses from class-specific log-uniform distributions. These inputs were shared across the Latent Trait, ATN, and ADBM models. Structural models were instead parameterised by a target connectance drawn from U(0.05, 0.3) where relevant. This allows one to isolate the effect of network-generating assumptions by holding constant (i) species richness, (ii) trait distributions (within trait-based models), and (iii) dynamical rules. Differences in structure and dynamics can therefore be attributed

directly to the generative model used to reconstruct trophic interactions.

Networks were generated using a rejection-sampling procedure to enforce comparable emergent properties. A network was retained only if its realised basal fraction and connectance fell within the target ranges (0.1–0.3 and 0.05–0.3, respectively). For trait-based models, networks were accepted only when all three models successfully generated valid networks from the same species pool. We obtained 100 valid replicates per model, with an upper bound on sampling attempts to prevent non-termination. MaxEnt networks were constructed by sampling joint in- and out-degree distributions consistent with a target number of links, followed by entropy-maximising rewiring under degree constraints. The Niche and Cascade networks were constructed using the target connectance and species richness.

2.2 Network structure

All networks were converted to directed interaction networks and analysed using a standardised pipeline. We quantified thirty structural properties across multiple scales (See table S1 for the full list), including connectance, trophic level, generality, vulnerability, clustering, centrality, trophic coherence, and path length. These metrics were used to characterise the structural signature of each model.

2.3 Network equilibrium and stability

To evaluate the dynamical consequences of different generative models, we simulated each network using the bio-energetic food web dynamic model implemented in `EcologicalNetworksDynamics.jl` (Lajaaiti et al., 2025). A key feature of the bio-energetic model is the allometric scaling of metabolism, growth and foraging rates with body mass (Brose et al., 2006). For trait-based models (ATN, ADBM, LTN), species body masses were taken from the generative networks and rescaled relative to the producer with the lowest body mass. For structural models (Niche, Cascade, MaxEnt, Random), which did not include body mass information, body masses were assigned from species trophic levels assuming a predator–prey body mass ratio of 100 (Lajaaiti et al., 2025).

For a consumer species i, biomass dynamics were determined by gains from feeding, losses to predators, and metabolic maintenance, which is expressed as:

$$\frac{dB_i}{dt} = \sum_{j \in \text{prey}(i)} e_{ij} B_i F_{ij} - \sum_{j \in \text{predators}(i)} B_j F_{ji} - x_i B_i$$

where B_i is the biomass of consumer i, e_{ij} is its assimilation efficiency when feeding on prey j, x_i is its metabolic rate scaled with body mass, and F_{ij} is the functional response of consumer i feeding on prey j, defined as

$$F_{ij} = \frac{\omega_i a_{ij} B_j^h}{M_i \left(1 + \sum_{k \in \text{prey}(i)} \omega_i a_{ik} h_{ik} B_k^h\right)}$$

where w_i is the equal preference of the consumer i feeding on its prey, $w_i = 1/\text{number of}$ prey species. a_{ij} is the attack rate, h_{ij} is the handling time, and M_i is the body mass of consumer i. Attack rates and handling times scaled with both consumer and prey body masses. The Hill exponent was set to $h = 2$ as a Type III functional response.

For a basal species j, biomass dynamics were determined by its logistic growth and losses to consumers, expressed as:

$$\frac{dB_j}{dt} = r_j G_j B_j - \sum_{i \in \text{predators}(j)} B_i F_{ij}$$

where r_j is the intrinsic growth rate and $G_j = 1 - \frac{B_j}{K_j}$ is the logistic growth modifier , with K_j is the carrying capacity. Both r_j and K_j scaled with species body mass.

All allometric constants used to parametrise x_i, a_{ij}, h_{ij}, r_j, and K_j were set to the default values provided by (Lajaaiti et al., 2025).

We first ran a burn-in simulation for each food web until a steady-state equilibrium was reached. Species with biomass below 10^{-6} were considered extinct, and basal species without consumers and consumers without prey were removed. We then simulated the remaining food web again to a steady-state equilibrium, yielding to the realised feasible network. We also recorded cases in which food webs failed to reach equilibrium.

Based on the realised networks and equilibrium biomasses, we calculated species persistence and the Shannon diversity of species biomasses. We extracted the consumption fluxes along individual trophic links and calculated their Gini coefficient (Altena et al., 2016), with higher values indicating more uneven distributions of fluxes. We further derived the Jacobian matrix, whose non-zero off-diagonal elements represent the pairwise interspecific

interaction strengths, comprising per capita top-down and bottom-up effects. We calculated the the skewness of interspecific interaction strengths to characterise the asymmetry of their distribution, with greater positive skewness reflecting many weak interactions and a few strong interactions (Berlow et al., 2004).

Finally we assessed local community stability using resilience and reactivity. Resilience was measured by the real part of the dominant eigenvalue of the Jacobian matrix, with more negative values indicating faster return to equilibrium following a small perturbation (Ruiter et al., 1995). Reactivity quantified the maximum instantaneous amplification of perturbations, with more positive values indicating a stronger tendency for some perturbations to initially move the system further away from equilibrium (Neubert & Caswell, 1997).

2.4 Statistical analysis

2.4.1 Phenotypic trajectory analysis of network structure

To quantify how ecological dynamics reshaped network topology, all structural metrics were recalculated following the dynamic simulations using the realised (post-extinction) network. These post-dynamics metrics were compared with those of the initial network using a phenotypic trajectory analysis (PTA) framework (Adams & Collyer, 2009), where each network was represented as a point in a multivariate topology space defined by the suite of structural network metrics (Table S1). Prior to analysis, all metrics were standardised (mean = 0, SD = 1) to remove differences in scale before constructing a common topology space using principal component analysis (PCA). Both pre- and post-dynamics networks were projected into this shared ordination, allowing each network to be represented by a trajectory from its initial to realised topology.

Trajectory vectors were calculated as the displacement between pre- and post-dynamics positions in the five-dimensional PCA space. Trajectory length was calculated as the Euclidean distance between the two states, providing a measure of the magnitude of topological change. Mean trajectories were calculated for each network reconstruction model centroid to summarise their overall direction of change. Finally, convergence among network topologies was assessed by calculating each network's Euclidean distance to the global post-dynamics centroid before and after simulation, with positive reductions in distance indicating convergence towards a common realised network topology.

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2.4.2 Structural equation model analysis

Ecological inferences drawn from generated food webs depend on how the choice of generative model shapes network structure and community dynamics. We therefore examined how variation in food web stability was associated with richness and topological distinctiveness, and whether these relationships differed among the seven generative models.

We used multi-group structural equation modelling (SEM) to quantify direct and indirect relationships linking richness and topology distance to resilience and reactivity through the Shannon diversity of species biomasses, the Gini coefficient of consumption fluxes , and the skewness in interaction strengths. Topology distance was calculated as the Euclidean distance from the pooled centroid in the principal component space of standardised topological metrics at equilibrium. The seven generative models shared a common pathway structure, with regression coefficients, intercepts, variances, and residual covariances estimated separately for each group. All variables were standardised using means and standard deviations calculated across the complete dataset, providing common units for coefficient comparisons. Models were fitted using robust maximum likelihood in lavaan (Rosseel, 2012).

We then decomposed the model-implied variance in each stability response into contributions associated with richness, residual topological variation, and residual variation in biomass, flux, and interaction properties. Each contribution was calculated by propagating its source variation through all direct and indirect pathways to stability. The richness contribution included its direct effect and all indirect effects through downstream variables. The topological contribution used the residual variance of topological distance after accounting for richness and its total downstream effect on stability. The biomass, flux, and interaction contribution jointly propagated the residual variation in biomass diversity, flux Gini, and interaction-strength skewness. These three contributions sum to the full SEM (R^2), with (1-R^2) representing unexplained variation. This decomposition was performed separately for each stability response and generative model. For example, the fraction of resilience variance attributed to richness was calculated as $\frac{\tau_{S,Y}^2 \operatorname{Var}(S)}{\operatorname{Var}(Y)}$, where S denotes richness, Y denotes the stability response, and $\tau_{S,Y}$ is the total effect of richness on resilience, comprising the direct path coefficient plus the sum of coefficient products

along all indirect pathways. The variances $Var(Y)$ refer to the fitted SEM within each generative model.

3 Results

3.1 Networks Converge in Structure

Projection of all networks into a common topology space revealed that initial and realised food web topologies occupied distinct regions of multivariate space (Figure 1 A). The first two principal components explained 50.7% of the total variation in network structure, with variation primarily associated with linkage density, chain length, and direct competition (S4) (Figure 1 B). While considerable separation among reconstruction models was evident in the initial topology space, post-dynamics networks exhibited greater overlap, suggesting that ecological dynamics altered structural differences among models.

[Figure 1 about here.]

All reconstruction models underwent measurable shifts in topology following dynamic simulations, although both the magnitude and direction of change differed among models (Figure 1 C). Centroid trajectories showed that all models moved broadly in the same direction, indicating that ecological dynamics reshaped network topology in a consistent manner. This can be seen when looking at the displacement of models along the various PC axes (Figure 1 D), where all models showed the greatest displacement along PC1.

Finally, realised networks showed evidence of convergence within topology space (Figure 1 E). The mean Euclidean distance to the global realised-network centroid decreased following dynamic simulations for all models, suggesting that ecological dynamics drove networks towards a common structural endpoint despite differences in initial topology.

3.2 Structure does not confer stability

With the exception of the Niche and Cascade models not all networks reached an equilibrium biomass (Figure 2). The Random networks had the lowest percentage of networks that reached an equilibrium state, with most of the networks collapsing *i.e.,* species went extinct. With the two structural networks (Random and MaxEnt) we see that the larger the initial network is the more of them fail to reach an equilibrium state. Conversely the three body size models (ADBM, ATN, and LTM) show the opposite relationship (smaller networks tend to fail). Interestingly we primarily see that networks fail due to extinction

of the community as opposed to the network not being able to reach an equilibrium state.

[Figure 2 about here.]

Interestingly if we plot the 'failed' networks into the PCA space (Figure 3, darker points) these networks do not differ obviously in structure with the 'successful' networks. What is perhaps notable though is that we do begin to see some deviation in this observation for the larger networks (right column, Figure 3), with these networks typically being on the 'end range' of the pre structural space.

[Figure 3 about here.]

[Figure 4 about here.]

3.3 Structure-stability relationships varied despite structural convergence

Despite convergence in food web structure, relationships linking structure to stability varied among generative models (Figure 5). After accounting for richness, residual topological distance accounted for 0-7% of total variation in resilience and 0-13% in reactivity. Its contribution to reactivity was greatest in Niche (10%) and Cascade (13%). Residual variation in biomass diversity, flux unevenness, and interaction strength skewness jointly made the largest contribution to resilience variation in every model, accounting for 12-40% of total variance, compared with 3-16% attributed to direct and indirect pathways from richness. This combined contribution was particularly large in ADBM (40%) and ATN (28%), whereas richness and residual topology together accounted for 18% in each model.

Among the direct paths to stability, greater flux unevenness was associated with slower recovery in six models. Greater interaction strength skewness was significantly associated with faster recovery in ADBM, Niche, Cascade, and MaxEnt, however its relationship with reactivity changed direction among models: positive in ADBM, ATN, and LTM, and negative in Niche and Cascade. Thus, even where topology converged, the pathways linking biomass, flux, and interaction properties to stability remained dependent on the generative model.

[Figure 5 about here.]

4 Discussion

4.1 Ecological dynamics constrain food web structure

One of the clearest results of this study is that ecological dynamics consistently reduced structural differences among food web reconstruction frameworks. Networks generated from fundamentally different structural, statistical, and trait-based assumptions began in distinct regions of multivariate topology space, yet their realised counterparts occupied a much narrower region following bioenergetic simulations. This convergence indicates that ecological dynamics impose strong constraints on viable food web architecture, regardless of how the initial network is assembled.

Importantly, this convergence emerged without any opportunity for networks to rewire. Species interactions remained fixed throughout the simulations, meaning that all structural change resulted from species extinctions and the subsequent removal of trophic interactions. The realised networks therefore represent a filtered subset of the original architectures rather than newly assembled communities. In this sense, bioenergetic dynamics act as a selection process on network structure. Interactions that cannot be maintained under energetic and demographic constraints are systematically removed, while those compatible with persistence remain. This interpretation aligns closely with previous work demonstrating that secondary extinctions reshape food web structure through dynamic processes that cannot be captured by static robustness analyses alone (Curtsdotter et al., 2011).

The dominant axis of structural change was associated with reductions in metrics describing indirect interactions or feedback loops (such as omnivory). This interpretation is consistent with empirical studies showing that food webs exhibit conserved local interaction structure across ecosystems despite substantial differences in species composition, suggesting that persistent communities occupy a constrained distribution of smaller motifs (Stouffer et al., 2007; Stouffer & Bascompte, 2010), and underscores the role of loops in driving network resilience (Li et al., 2021). Our simulations extend this idea by demonstrating that similar structural organisation can emerge through extinction-driven filtering, even when networks originate from contrasting generative assumptions. Moreover, this extinction-driven pruning agrees with the observation that stable food webs are enriched for interaction configurations that persist under ecological dynamics, whereas structurally unstable configurations are selectively removed (Borrelli, 2015).

The behaviour of the Random networks further illustrates the existence of a constrained structural domain for persistence. Randomly assembled networks were the least likely to reach equilibrium (Figure 2), particularly as species richness increased, yet those that did survive underwent relatively little structural reorganisation (Figure 1 D). Rather than converging through extensive pruning, these surviving networks appear to have been generated close to a dynamically feasible region of topology space by chance. This result echoes classic stability theory, which predicts that unconstrained random interaction matrices are overwhelmingly unstable unless they possess particular structural properties (Allesina & Tang, 2012; May, 1974). Ecologically informed reconstruction models do not guarantee persistence, but they substantially increase the probability of generating networks within this 'feasible region' because their assembly rules embed ecological constraints absent from purely random topology (Dunne et al., 2002).

Together, these results suggest that convergence in topology is not evidence that generative models are interchangeable. Instead, ecological dynamics compress a diverse set of initial architectures into a narrower subset of structures compatible with persistence. The important question therefore becomes whether this shared realised topology also produces shared ecological dynamics, or whether the assumptions embedded in the reconstruction framework continue to influence stability despite structural convergence.

4.2 Structural convergence does not imply dynamical equivalence

Although ecological dynamics drove networks towards a shared region of topology space, this convergence did not produce equivalent predictions of community stability (Figure 5 and S1). Instead, we observed a clear separation between coarse dynamic properties, which were largely explained by realised species richness, and local dynamical properties, which remained strongly dependent on the original reconstruction framework (Figure 5 and S2). Structural convergence therefore represents only one dimension of ecological similarity; the dynamical consequences of those structures retain a signature of how the network was assembled.

This distinction is most apparent when comparing network-level and local stability metrics. Shannon diversity of biomasses, interaction strength skewness, species persistence, and the Gini coefficient of consumption fluxes were explained predominantly by equilibrium richness rather than model identity. This agrees with longstanding ecological theory showing that many aggregate properties of communities emerge from biodiversity itself

through biomass partitioning, portfolio effects, and the distribution of energy across trophic levels (Loreau & Mazancourt, 2013; Yodzis, 1981). Once ecological dynamics had removed species and reduced networks to similar realised sizes, these macroscopic properties became remarkably insensitive to the initial reconstruction framework.

Local stability, however, behaves differently. Reactivity was explained almost entirely by the generative model, while resilience remained strongly influenced by model identity even after accounting for realised richness. These metrics quantify how perturbations propagate through communities rather than simply whether communities persist, making them inherently sensitive to the organisation of interaction strengths within the Jacobian matrix (Neubert & Caswell, 1997; Tang et al., 2014). Two networks with similar connectance, trophic structure, and species richness can therefore exhibit fundamentally different transient responses if the underlying energetic constraints that generated their interactions differ.

The contrast between structural and trait-based models illustrates this point. Niche and Cascade networks consistently reached equilibrium, but this should not be interpreted as evidence that they provide the most biologically realistic representation of ecological persistence. Rather, these models inherit body masses only after network construction through a fixed predator-prey mass ratio, meaning that energetic structure is imposed *post hoc* to satisfy the assumptions of the bioenergetic model (Delmas et al., 2017). In contrast, ADBM, ATN, and latent trait networks propagate body mass distributions directly from the reconstruction process into the dynamical model. The resulting interaction strengths, attack rates, handling times, and metabolic rates remain coupled through the biological assumptions embedded in each framework (Brose et al., 2006; Petchey et al., 2008; Schneider et al., 2016).

This difference helps explain the apparent paradox in our results. Trait-based models often supported fewer persistent species than structural models, yet they generated markedly different resilience and reactivity profiles. In particular, latent trait networks combined relatively high resilience with the strongest transient amplification of perturbations, while ADBM networks recovered rapidly despite lower overall persistence. This decoupling of asymptotic stability from transient dynamics has been recognised as a defining property of complex ecological communities where interaction strengths are structured rather than random (Neubert & Caswell, 1997; Tang et al., 2014). Consequently, reproducing global food web topology is not sufficient for reproducing ecological dynamics.

Our results suggest that structural convergence should not be interpreted as evidence that reconstruction frameworks become interchangeable after ecological dynamics. Instead, extinction-driven filtering removes many topological differences while preserving historically contingent differences in how energy flows through the community. The realised network may look similar, but its response to perturbation still reflects the assumptions embedded in the model used to generate it.

4.3 What makes a good food web model?

The Niche model transformed food web ecology by demonstrating that simple assembly rules can reproduce the large-scale architecture of empirical trophic networks (Williams & Martinez, 2000). Since then, structural generative models have become the default starting point for studies of robustness, extinction cascades, and ecosystem stability (Allesina & Tang, 2012; Curtsdotter et al., 2011). Our results do not challenge the value of these models for describing food web topology. Instead, they demonstrate that reproducing network structure is not equivalent to reproducing ecological dynamics.

The central implication of this study is that food web reconstruction frameworks should be viewed as an embedding of assumptions rather than interchangeable Null models. Each reconstruction framework encodes assumptions about the processes that generate trophic interactions, whether these assumptions arise from niche ordering, statistical constraints, body size scaling, or optimal foraging. Those assumptions influence the interaction strengths, energetic organisation, and transient dynamics that emerge once networks are embedded within a common dynamical framework. Ecological dynamics subsequently filter these initial structures towards a shared region of viable topology, but they do not erase the signature of how those networks were constructed.

This distinction helps reconcile an apparent tension in the food web literature. Structural models have proven remarkably successful at reproducing global network statistics across ecosystems, while mechanistic models have been developed to explain the biological processes that generate those same interactions (Banville et al., 2023; Brose et al., 2006; Petchey et al., 2008). Our results suggest that these approaches answer different ecological questions rather than competing to identify a single 'best' food web model. For questions concerning broad topological organisation, biodiversity patterns, or realised species richness, structural and statistical models may provide appropriate and computationally efficient approximations. In contrast, questions concerning transient dynamics, perturbation

responses, resilience, or extinction cascades require reconstruction frameworks that preserve the biological constraints governing interaction strengths and energy flow (Delmas et al., 2017; Poisot et al., 2016; Schneider et al., 2016).

More broadly, these findings support a shift in how food web reconstruction is interpreted. Rather than asking whether one generative model best represents ecological communities, we should ask whether the assumptions embedded within a reconstruction framework are appropriate for the ecological inference being made. As ecological networks are increasingly reconstructed across space, time, and future environmental scenarios (Strydom et al., 2021; Strydom, Dunhill, et al., 2026), explicitly matching reconstruction models to inferential goals will become as important as matching dynamical models to ecological processes.

Ultimately, not everything is a niche model. Different reconstruction frameworks can converge on similar realised network architectures while retaining fundamentally different predictions about how ecological communities respond to perturbation. The choice of generative model is therefore not simply a technical decision about network construction. It is an ecological hypothesis about how communities are assembled and how they persist.

4.4 Implications for linking form to function

Food webs have long been viewed as a framework for linking community structure to ecosystem functioning, connecting patterns of species diversity and trophic interactions with the movement of energy and materials through ecological communities (Thompson et al., 2012). However, establishing how reliably food-web structure can predict ecosystem functioning remains a central challenge. In particular, binary representations of food-web structure describe who interacts with whom, but do not necessarily capture the strength or functional consequences of those interactions. The importance of interaction strength for understanding population dynamics and community-level processes has been widely recognised, with the arrangement of strong and weak interactions often being as important as the presence or absence of links themselves (Berlow et al., 2004). Similarly, attempts to explicitly connect food-web structure to different ecological functions have shown that structures associated with resilience need not be those associated with energy throughput (Yen et al., 2016). Together, this literature highlights that there is no straightforward or universal mapping between network topology and ecosystem function.

Our results add to this discussion by showing that the relationship between form and function is complicated not only by the distinction between binary links and quantitative

interaction strengths, but also by the assumptions underlying food-web reconstruction. Despite convergence in realised topology, our models retained differences in biomass distributions, energy fluxes and local stability. This suggests that structural similarity does not necessarily indicate that networks retain equivalent ecological information. In our comparison, different reconstruction frameworks encode different assumptions about the factors governing trophic interactions, including trophic ordering, organismal traits and energetic constraints, or network-level properties. These assumptions can become consequential when networks are used to predict ecosystem dynamics, even where their realised topologies appear similar.

An important implication is that the structural properties most useful for predicting ecosystem behaviour may depend on the particular function or dynamical response being considered. Broad community properties may be captured reasonably well by relatively coarse descriptions of network structure, whereas predicting energy transfer or responses to perturbation may require information about biomass distributions, interaction strengths and the mechanisms governing feeding interactions. This distinction also emphasises that ecosystem functioning and stability should not be treated as interchangeable outcomes: a network structure that supports one property need not maximise or predict another (Yen et al., 2016). Thus, rather than asking whether food-web structure predicts function in general, a more productive question may be which aspects of structure, and which underlying ecological processes, are necessary to predict particular components of ecosystem functioning.

Ultimately, our findings suggest that food-web models should be evaluated not only on their ability to reproduce observed network topology, but also on whether they retain the ecological information needed to predict the functional and dynamical properties of interest. Structural agreement provides an important basis for comparison, but it is not sufficient evidence of functional equivalence. Progress in linking form to function will therefore depend on combining structural descriptions of ecological networks with quantitative information on interactions and the biological mechanisms that shape them.

5 Conclusion

Our results demonstrate that ecological dynamics can drive food webs generated from contrasting reconstruction frameworks towards similar realised structures, even in the absence of rewiring. However, this structural convergence does not translate into convergence in all

aspects of ecosystem dynamics. While some broad community and aggregate properties were associated primarily with realised species richness, local stability properties remained strongly dependent on the reconstruction framework. These findings highlight that similar food web structures can retain different dynamical properties, reflecting the ecological assumptions and information encoded in the models from which they originate.

More broadly, our study highlights an important distinction between reproducing the form of ecological networks and capturing the processes that determine their function. Different reconstruction frameworks encode different assumptions about the factors governing interactions, whether these are trophic ordering, organismal traits and energetic constraints, or network-level properties. These assumptions shape not only the structures that models generate, but also the ecological information carried into subsequent dynamical predictions. Consequently, evaluating food web models solely on their ability to reproduce observed topology may provide an incomplete assessment of their ecological value. Understanding when and why different models produce similar structures but different dynamics is essential for determining which representations of food web structure are appropriate for predicting particular aspects of ecosystem functioning.

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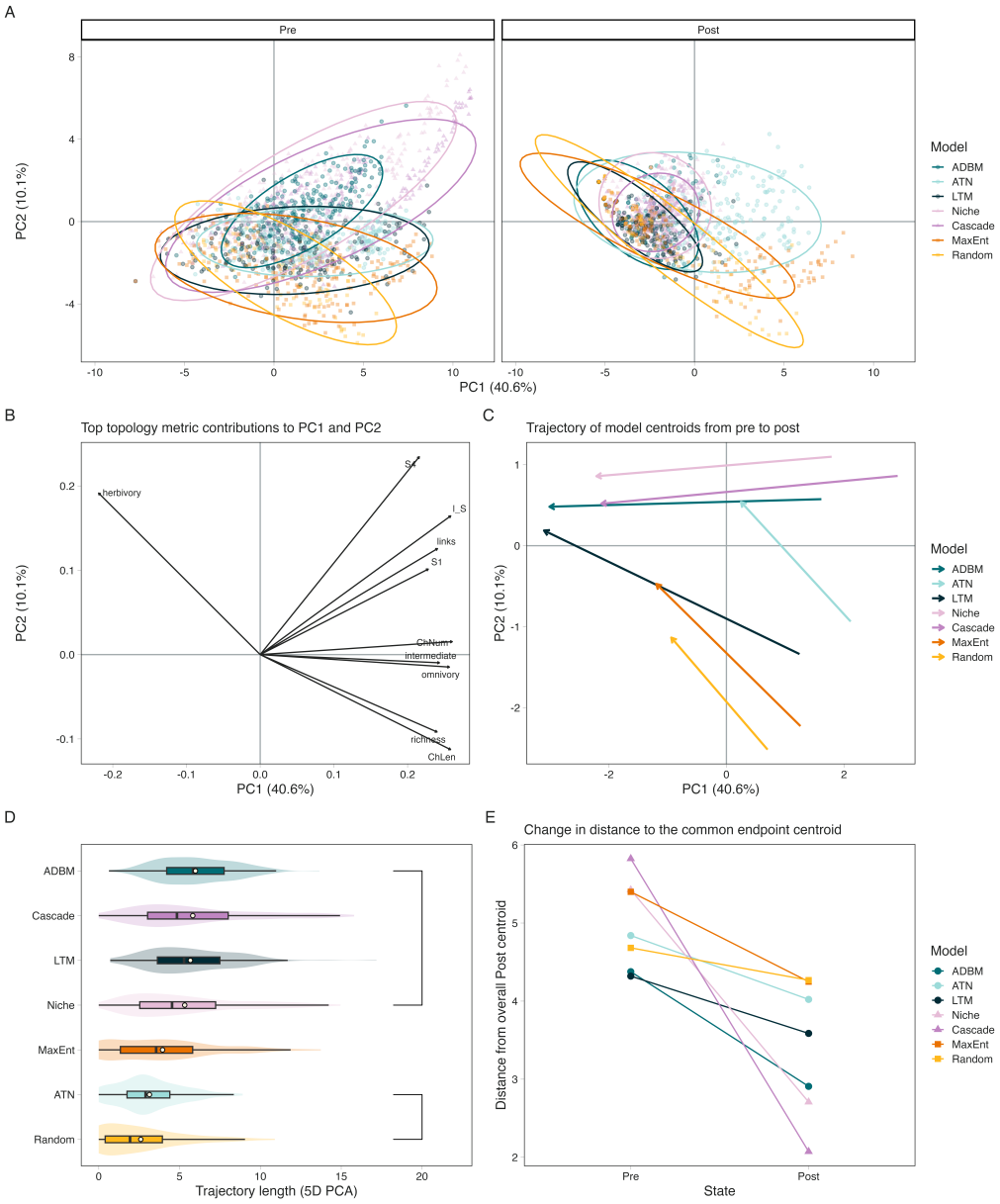


Figure 1: Ecological dynamics reshape food web topology within a shared multivariate topology space. Network structural metrics were standardised and summarised using principal component analysis, with each network represented by its position before and after dynamic simulations. (A) PCA of networks before and after dynamic simulations. Ellipses cover 90% of points (B) Principal component loadings showing the contribution of the top ten individual network metrics to topology space (C) Centroid trajectories for each reconstruction model, illustrating the direction and magnitude of topological change between initial and realised networks. (D) Displacement of model centroids as trajectory lengths over the first five principal component axes, points represent mean length (E) Mean Euclidean distance of networks to the centroid before and after simulations, providing a measure of convergence in network topology.

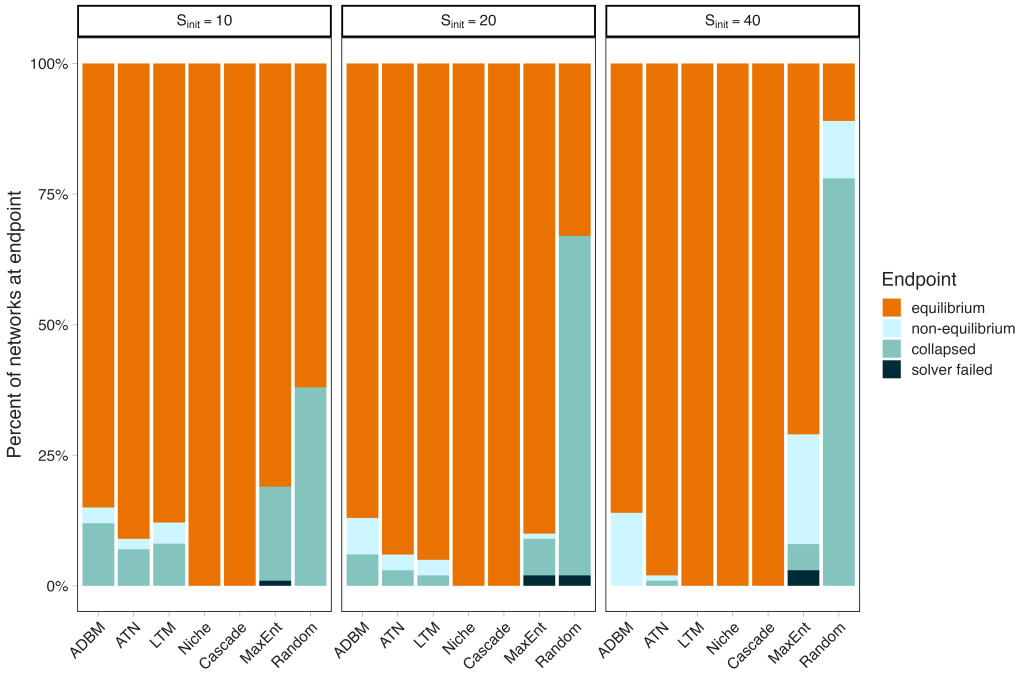


Figure 2: End points of networks indicating the percentage of networks that reached an equilibrium biomass (orange) and if they did not (shades of blue) why not. These are faceted by the initial starting richness for each network

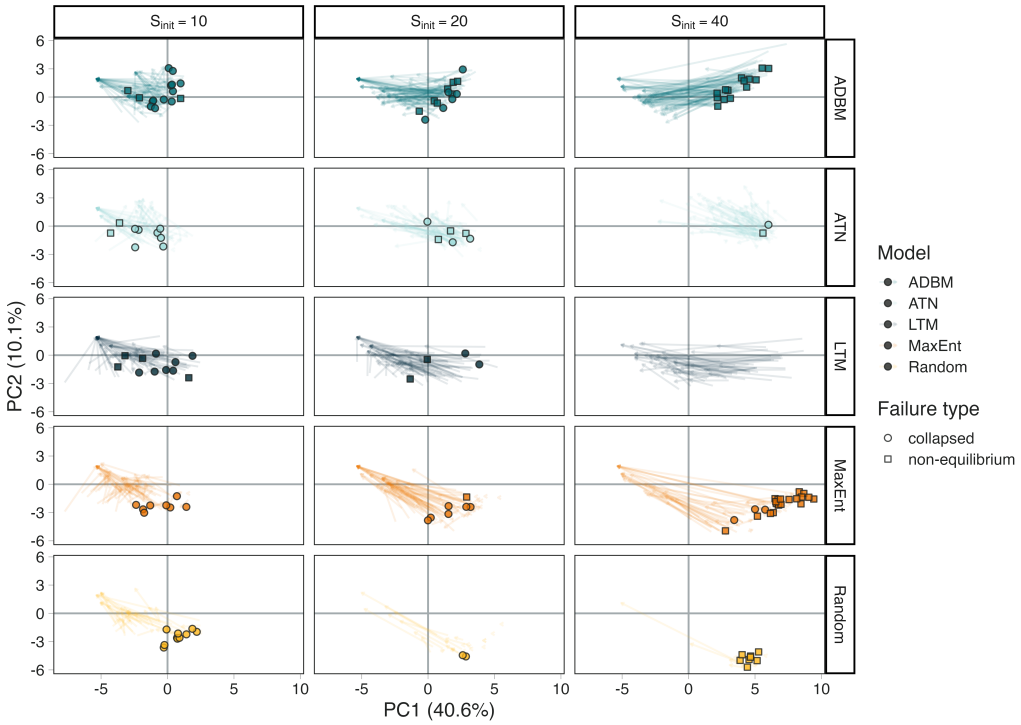


Figure 3: Arrows indicate trajectory of individual networks. Points are networks that failed to reach an equilibrium state that have been predicted into the PCA space. Pale arrows show the trajectory of each individual network, dark arrows the mean trajectory. Note we do not plot networks that failed due to solver issues, additionally we omit the Niche and Cascade models as they have no failed networks

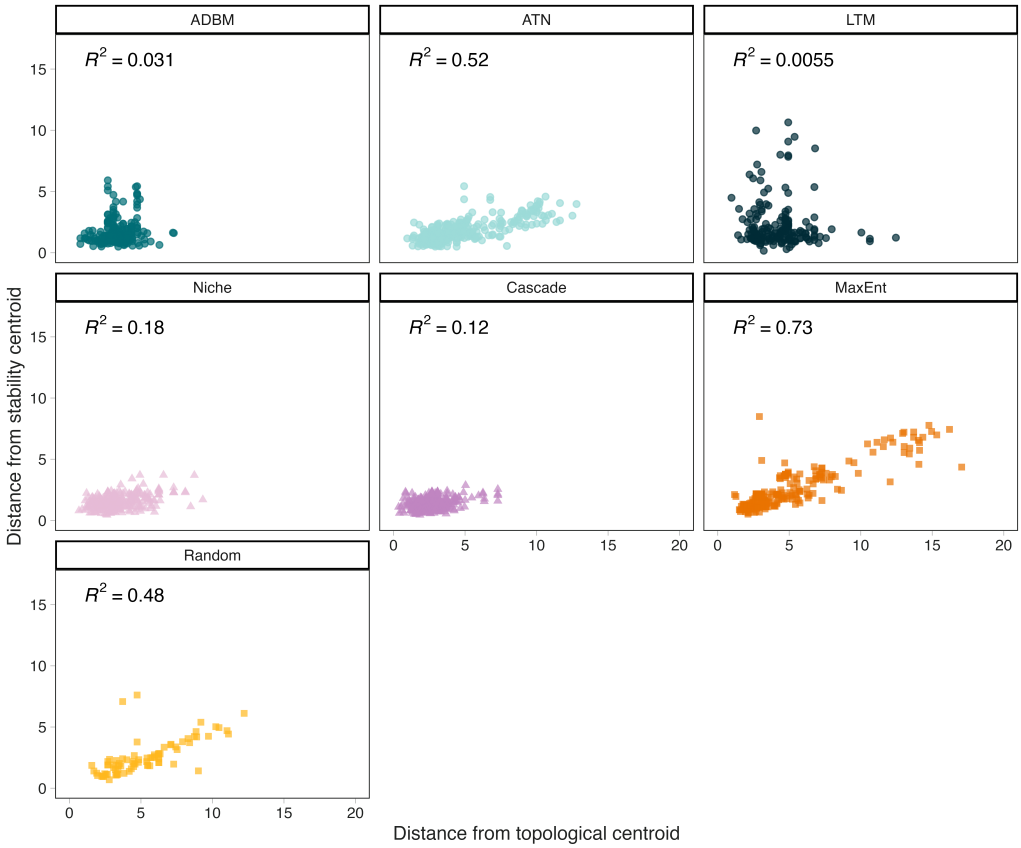


Figure 4: words

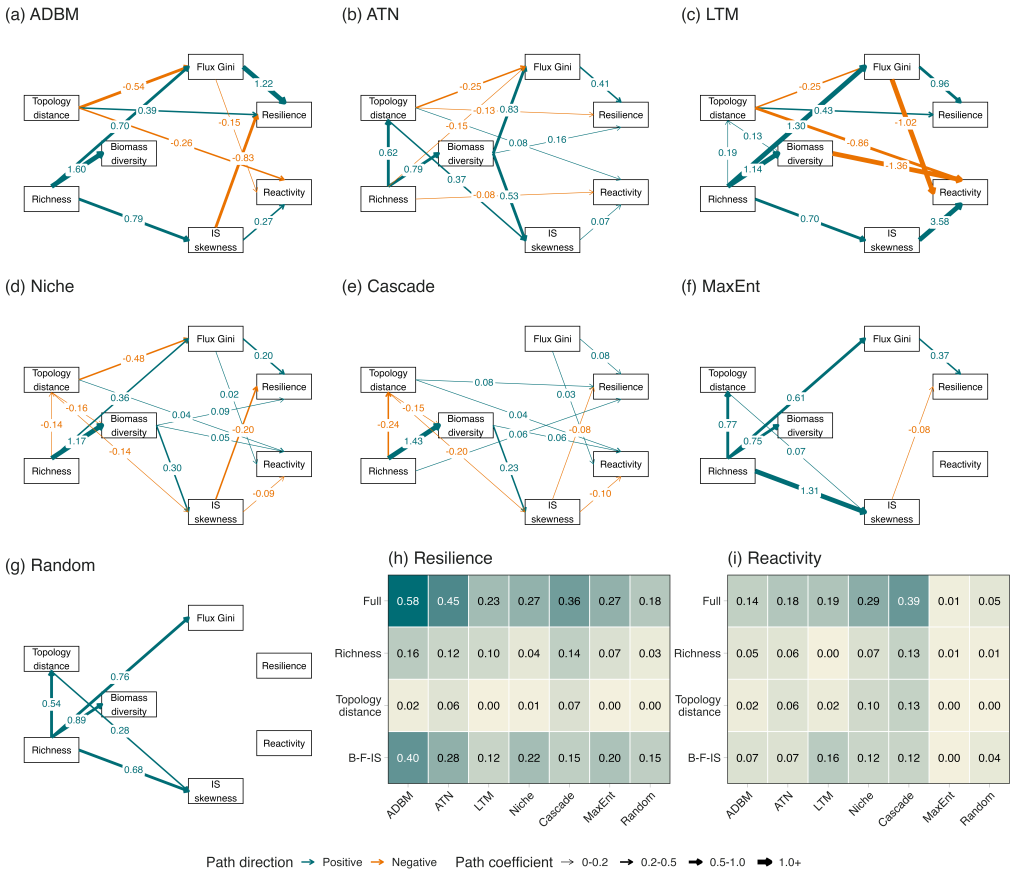


Figure 5: Structural equation models linking food web structure to local community stability across seven generative models. (a-g) Significant paths ($p < 0.05$) for ADBM, ATN, LTM, Niche, Cascade, MaxEnt, and Random. Blue and red arrows indicate positive and negative relationships, respectively; Topology distance measures distance from the pooled centroid in topological principal component space. Flux Gini and IS skewness denote the Gini coefficient of consumption fluxes and skewness of absolute interaction strengths. Higher resilience values indicate slower recovery; higher reactivity indicates greater initial perturbation amplification. (h, i) Fractions of total variance in resilience and reactivity explained by the full SEM and attributed to richness, residual topological variation, and residual biomass, flux, and interaction variation (B-F-IS), including their downstream pathways.